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Affective Narrative Application to Games Involving Narrative Choices

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Abstract

This research paper investigates affective narrative in digital games involving narrative choices. Specifically, it examines how player perspective influences decision making, how tension between player agency and rigid narrative resolution creates dissonance and immersion breakdown, and how players' decisions and emotional stability shape the narrative and psychological impact of the experience.

The findings of this study aim to inform industry specialists, academic scholars, and game developers about the importance of interactive narrative. The research was conducted individually through separate research questions, with results combined into a cohesive analysis in the format of a Kappa paper. A game artefact was developed to support data collection.

The research produced findings consistent with existing studies; however, the small sample size limits the ability to generalize the results and indicates a need for further research. The study suggests that subverting player narrative agency is most effective when agency has first been established through both narrative and technical design. Additionally, the "Guilt-o-meter" influenced players' moral judgments and interpretations of characters, with stronger feelings of guilt and hesitation occurring among players who formed emotional connections with the NPCs.

Keywords: Branching dialogue, character attachments, emotional engagement, illusion of choice, research through design, video games

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1 Introduction

Guilt-o-meter is a first-person narrative adventure game in which players assume the role of a floor manager working within a morally questionable corporation. The player's primary task is to identify and terminate "guilty" employees using the company-issued device known as the Guilt-o-Meter. However, as the narrative unfolds, it becomes evident that the device does not measure guilt at all; instead, it targets employees with the highest salaries as part of a corporate downsizing strategy. Through this premise, the game challenges players' trust in systems of authority and raises questions about morality, manipulation, and responsibility within interactive narratives, which informs the research question: How can a Design team apply affective narrative to games involving narrative choices?

The overarching aim of this research project was to investigate the application of narrative choices in games, with a particular focus on how unreliable feedback mechanisms influence player agency and the overall narrative experience. This research is significant because it provides insight into how players interpret and respond to deceptive narrative systems, as well as how these systems affect immersion, emotional engagement, and decision-making processes in digital games.

This thesis follows a KAPPA synthesis format and consists of a collaborative framework accompanied by three individual research papers. The project was developed by a team of three researchers, each examining a distinct aspect of the Guilt-o-meter experience.

Paper A, *Perspective*, explores how player perspective influences decision-making and shapes the player's interpretation of narrative events. Paper B, *The Illusion of Choice*, investigates narrative dissonance and the breakdown of immersion caused by the tension between perceived player agency and predetermined narrative outcomes. Finally, Paper C, *Moral Agency*, examines how player decisions and emotional responses influence the narrative trajectory and psychological impact of the game experience, demonstrating how digital games can function not only as entertainment but also as spaces for exploring human values and ethical decision-making.

While each paper focuses on a distinct analytical lens, they are intentionally interlinked through a shared examination of how player agency is constructed, challenged, and destabilized within the game. Together, they form a layered understanding of the *Guilt-o-*

meter experience: Paper A establishes how perspective frames player interpretation, Paper B interrogates how that interpretation is disrupted through illusory choice structures, and Paper C explores the resulting emotional and moral consequences. In combination, the three studies provide a cohesive analysis of how narrative design can shape, and undermine, the sense of agency in interactive storytelling.

2 Background

Guilt-o-meter builds upon the "Bureaucracy Simulator" genre, drawing on the inspiration of *Papers, Please* (Pope, 2013), *Not Tonight* (PanicBarn, 2018), and *Beholder* (Warm Lamp Games, 2016). Similarly to these titles, this project explores the moral erosion of an individual working withing a restrictive system. However, where *Papers, Please* (Pope, 2013) offers multiple endings based on player performance, *Guilt-o-meter* explores the design space of narrative inevitability. The research team draws inspiration from *The Stanley Parable* (Wreden, 2013) regarding its subversion of player expectations and the use of an unreliable guide.

This research is informed by the following key theoretical frameworks:

1. Meaningful and Inconsequential Choices (Eriksson Turtiainen, 2018). The thesis investigates how different types of story choices impact a player's sense of agency. In it, the author distinguishes between "meaningful choices", which tangibly alter the game state, and "inconsequential choices", which do not change the objective outcome. The Research Team's game artefact utilizes this framework by leaning heavily on inconsequential choices within its fixed narrative, deliberately provoking deep moral engagement and risking player frustration through the narrative inevitability.
2. Illusion of Choice (Alborov, 2017). Alborov examines how games can create a strong feeling of agency through the illusion of branching outcomes, even when the underlying narrative remains tightly controlled. The article argues that this

illusion is often more sustainable than fully diverging storylines. It also warns that the illusion of choice collapses into “bitter disappointment” once players realize their choices lack real consequence. The research team’s game artefact engages directly with this tension by building up perceived agency through dialogue and then confronting players with a rigid narrative resolution that exposes the limits of that illusion.

3. Beads on a String (Kiander, 2019). Kiander describes the “beads on a string" structure, where local segments of play (the beads) may branch and vary, while the overarching narrative string always pulls the story back to a shared path. This design can offer short-term agency and replay value as long as each bead feels mechanically and narratively coherent, even if long-term outcomes converge. The research team’s game artefact adopts this structure by using branching dialogue to create locally varied interactions, then funneling every narrative path into a single rigid narrative resolution that intentionally tests how visible the underlying string becomes to players.
4. Character Attachment (CA) defined as;

   > *we define CA as an individual’s feelings of (a) friendship and (b) identification with a video game character when an individual (c) is willing to suspend disbelief, (d) feels responsible for the game character, and (e) feels in control of the game character’s actions.* (Lewis et al., 2008).

   Will be a crucial framework for both gathering and interpreting data since it provides 5 concrete criteria to evaluate. Furthermore, by using this framework as a guide the results are more likely to be compatible and comparable to other research allowing for parallels and comparisons to be made.

5. Qin, et al. (2009). Measuring Player Immersion in the Computer Game Narrative. Proposes a framework for measuring immersion by tracking 7 variables; Curiosity, Comprehension, Challenge / Skills, Empathy, Concentration, Control and Familiarity. This model will be used to measure the level of immersion playtesters are experiencing.
6. Liu, Z. & The University of New South Wales, Sydney, Australia. (2025). The impact of moral choices in games on players' moral values. In Proceedings of the 6th International Conference on Educational Innovation and Philosophical Inquiries: The paper defines games as "moral laboratories" where "ambiguous decisions" trigger self-examination. In Guilt-o-meter mirrors this by placing players in a role where the player must make decisions to fire people. Using unreliable feedback to fire high earners creates a "moral conflict" that transforms your game into a space for "moral reflection"
7. A History of Moral Choice in Games (Gibbons, n.d.). This work looks at how moral choices in games have changed over time, from simple good vs bad systems to more complex designs that focus on player interpretation and emotional impact. It explains that earlier games often used clear rewards and punishments, while newer games focus more on uncertainty and personal reflection. This is relevant to *Guilt-o-meter* because it supports the idea of moving away from simple morality systems instead focusing on emotional feedback and how players understand and react to their choices.

In this study, the research team agreed that to explore "affective narrative" three perspectives needed to be observed: "Player Perspective", "Narrative Agency", and "Moral Responsibility". Through these perspectives, the research team intended to generate insights into "affective narrative"

To inform “Narrative Agency” the study focuses on “meaningful choices” that are defined as decisions that tangibly change the game state, and “inconsequential choices” as those that leave the objective outcome unchanged but still let the players express and reflect on their character (Eriksson Turtiainen, 2018). Using this framework, the research examines how both types of choices contribute to perceived agency when combined with the illusion of choice (Alborov, 2017) and a beads-on-a-string narrative structure (Kiander, 2019).

To observe “Player Perspective” the framework of Character Attachment (CA), (Lewis et al., 2008) is the baseline theoretical framework used in this paper to explore a player’s attachment to their in-game avatar. The 7-Factor model (Qin et al., 2009) is the secondary framework used. These two frameworks are similar in that they track specific traits and emotions. By combining them and modifying them to remove certain traits that are not applicable to the game the research team can track players and gain greater insight into overarching patterns that would be difficult to detect otherwise.

Liu (2025) and Gibbons (n.d.) trace game design shift from predictable "good vs. evil" rewards toward "moral laboratories" fueled by ambiguity and emotional reflection. Guilt-o-meter embodies this evolution. By forcing players to terminate employees based on flawed, unreliable data, you disrupt the typical corporate optimization mindset. This structure ties directly to moral responsibility: because the system lacks clear ethical guidance, players cannot offload the blame onto mechanics. They must personally bear the psychological weight of their choices. Ultimately, the research uses engineered guilt to transform an automated gameplay loop into a profound, self-examining test of personal accountability.

Together, these theoretical frameworks use *Guilt-o-meter* as both a game artefact and a research device for examining how “player perspective”, “narrative agency”, and “moral responsibility” shape player experience. By using “Player Perspective” to inform design decisions combined with the illusion of meaningful choices and a deliberately fixed outcome,

the project investigates how players emotionally and ethically engage with systems that deny genuine control. The selected frameworks therefore provide both the conceptual foundation for game design and the analytical tools needed to evaluate player responses during testing. The following section outlines how these theories are translated into the practical design and methodological structure of the project.

3 Game Overview

Guilt-o-meter is a first-person detective mystery game set in an office and made for PC. The main mechanic is talking to employees and using the "Guilt-o-meter", a tool that ostensibly shows if a character is lying. The game is the main research tool used to study how players make moral choices, how they deal with responsibility, and how they connect with the player character.

This is a new project made for this research, not a continuation of an earlier game. It was created to study how players react to moral choices and emotional feedback in a controlled game setting. The game was built step by step, and changes were made during development based on feedback from playtests. This helped improve the dialogue system and the *Guilt-o-meter* over time.

The design focuses on three main ideas: system control over people, moral uncertainty, and corporate satire. The game takes place in a dark office setting, but it also has some humor. It is inspired by *Jazzpunk* (Necrophone Games, 2014). The office is meant to feel familiar but also strange, to match the themes of control and confusion.

The game uses dialogue where players can choose different responses, but the story always ends in the same way. This allows the research team see how players use small "micro-choices" to express themselves, even when they cannot change the final outcome. It also shows how players try to make their own meaning from limited choices.

The "Guilt-o-meter" is an important part of the game. It gives feedback about characters, but it is not always correct. This lets the researchers see when players trust the system and when they ignore it and trust their own judgment instead. It also helps show how much players depend on game systems when making decisions, even when those systems might be wrong.

The game is used to study how players react to pressure, unclear information, and fixed outcomes. It creates a controlled space where players' behavior can be observed in a clear way. Players are placed in situations where they must decide who to fire, even when they do not have full information. This creates situations where they must think carefully about their choices.

The game does not give clear rewards or punishments for moral choices. Instead, it focuses on unclear situations where players must decide for themselves what feels right. This helps show how people think about responsibility when the game does not guide them. It also

helps to see if players still feel guilt or responsibility when the game does not react in a clear or traditional way.

The environment is also designed to support research. The office setting creates a simple but controlled space where players can focus on dialogue and decision-making without too many distractions. The characters are written in a way that makes their intentions unclear, so players must interpret their behavior instead of being told what is right or wrong.

Overall, Guilt-o-meter is both a game and a research tool. It is used to study how story systems affect player choices, how emotional feedback changes decisions, and how players understand responsibility in games.

4 Research Design

This study follows a Research through Design (RtD) methodology, in which knowledge is produced through the iterative creation, testing, and refinement of a playable game artefact titled *Guilt-o-meter*. Rather than separating design and analysis into distinct phases, RtD integrates both, allowing insights to emerge directly from the act of designing and evaluating the system. The central aim of this approach is to investigate how narrative structure and emotional feedback systems influence player moral decision-making, particularly in constrained interactive environments.

The project is grounded in iterative design and reflective practice, where each development cycle is informed by findings from playtesting sessions. The prototype was developed in Unreal Engine, enabling the implementation of a first-person interactive office environment, dialogue systems, and the central "Guilt-o-meter" mechanic, which provides real-time emotional feedback during player decision-making.

Each iteration of development was evaluated through structured playtesting, and insights from these sessions directly informed adjustments to both technical systems and narrative structure. The research into "affective narrative" is informed by the three perspectives: meaningful versus inconsequential choices, character attachment, and moral responsibility in gameplay contexts. These frameworks collectively inform both the design of the artefact and the interpretation of player responses during the semi-structured interview.

"Player Perspective" was used to measure player immersion as well as to standardize the measurement of player immersion. Being a modified version of the (CA) (Lewis et al., 2008) and 7-factor model (Qin et al., 2009) it focused on specific traits and emotions which were translated into data by analyzing the semi-structured interviews. Each category had strict requirements limited to either a specific word or its derivative. Alternatively, an affirmative answer to a direct question would be accepted, for example, "I felt very immersed" would count as "immersion", while "I felt enveloped by the story" would not count as "immersion". Analysis is not limited to this method, however, it is useful to detect overarching trends and patterns.

To inform the "narrative agency", the research team designed the narrative structure of the game artefact to be deliberately player choice agnostic during the final narrative resolution. In line with Eriksson Turtiainen's (2018) focus on how story choices shape a

player's sense of control, the branching dialogue system presented a highly locally varied option that invite role-play and made it feel as if the player's input matters, even though all the narrative paths funnel into the same rigid narrative resolution building on the illusion of choice (Alborov, 2017). To gather insights into how players experienced this perceived agency within a beads-on-a-string structure (Kiander, 2019), the post-playtest interviews included targeted questions about the presented dialogue, the dialogue choices, and the overall narrative.

The research is guided by the perspective of moral responsibility in gameplay contexts. Z. Liu (2025) describes games as "moral laboratories" where difficult and ambiguous decisions encourage self-reflection. This connects to Guilt-o-meter, where players must decide who to fire using unreliable feedback, creating feelings of guilt and ethical uncertainty. Similarly, Gibbons explains that modern moral choice systems focus less on clear rewards and punishments and more on emotional impact and personal interpretation. As a result, Guilt-o-meter emphasizes moral responsibility through player reflection rather than simple right-or-wrong outcomes.

The research team consisted of three members, working collaboratively in a small-scale development structure. Two members were responsible for programming and technical implementation within Unreal Engine, including dialogue systems, interaction mechanics, and the implementation of the "Guilt-o-meter." The third member acted as both designer and project manager, overseeing narrative design, visual direction, and overall project coordination.

The team followed a consensus-based decision-making model, where all major design changes were discussed collectively and evaluated based on playtest feedback rather than individual preference. This ensured that iteration was grounded in empirical player responses rather than subjective assumptions.

Communication and coordination were managed exclusively through Discord, which functioned as the central platform for meetings, task discussion, feedback exchange, and development updates. This lightweight workflow supported rapid iteration while maintaining continuous alignment across all team members.

The study included a total of six playtest sessions, consisting of one pilot test and five formal playtests. Participants were recruited through friends and personal networks, resulting in a convenience sampling approach appropriate for exploratory RtD research. While this

limits generalizability, it enables in-depth qualitative insight into player experience and emotional response.

Across all sessions, participants interacted with a short playable segment lasting approximately 5–10 minutes, featuring a functional office environment, dialogue interactions with non-player characters (NPCs), and the “Guilt-o-meter” system. This system visually represented emotional consequence feedback tied to player decisions, particularly in relation to firing in-game employees.

Data collection consisted of Recorded gameplay sessions, capturing player movement, dialogue choices, and interaction patterns, Semi-structured interviews, conducted immediately after gameplay to explore emotional responses, moral reasoning, and perceptions of the system. Informal follow-up discussions where additional clarification was needed.

Participants were informed in advance about the nature of the study and provided informed consent before participation. They were also informed that gameplay would be recorded and that participation was voluntary, with the option to withdraw at any stage.

The collected data was analyzed using a thematic analysis approach, focusing on recurring behavioral patterns and emotional responses across both gameplay footage and interview transcripts. Attention was given to hesitation during decision-making, expressions of moral discomfort, and reliance on the “Guilt-o-meter” as a decision-making aid.

Ethical considerations were integrated throughout the research process. All participants signed informed consent forms prior to participation, which clearly outlined the purpose of the study, data collection procedures, and recording methods. Participants were informed that all gameplay sessions would be recorded and anonymized.

Given the morally sensitive nature of the game content, participants were briefed beforehand on the themes of workplace dismissal, ethical decision-making, and emotional feedback systems. They were also informed that they could withdraw at any time without penalty.

All collected data, including gameplay recordings and interview transcripts, was stored securely and accessible only to the research team. No personally identifiable information was included in the final analysis or documentation, ensuring full participant anonymity and compliance with ethical research standards.

This research is part of a collaborative academic project in which each team member produces an individual paper based on a shared dataset derived from the six playtest sessions. While the dataset remains consistent across all papers, each researcher applies a distinct analytical lens.

The methodological connection between papers lies in the shared RtD process, unified prototype, and common data collection structure. However, each paper focuses on different theoretical interpretations, such as meaningful and inconsequential choices, perspective and moral agency, and emotional feedback systems in interactive narratives.

This shared foundation allows for complementary analysis, where behavioral observation, thematic interpretation, and emotional response analysis can be cross-referenced across multiple academic perspectives.

Despite limitations introduced by technical challenges and the inability to complete the second planned iterative playtest cycle, this study maintains methodological rigor through structured RtD practice, systematic data collection, and triangulated qualitative analysis. By combining iterative design with established theoretical frameworks, the research ensures that findings are grounded in both empirical observation and game studies theory

5 Summary of Results

Paper A

Players were immersed in the game and behaved in accordance with what the literature predicted. Meaning that players for the most part played “as themselves” with notable outliers role-playing or “meta gaming”. Examples of these behaviors include parsing through every dialogue option to see them all or intentionally choosing evil or cruel options just to see what will happen. The results were mostly positive, however, due to the limited sample size and the sampling method (convenience sampling) the results cannot be generalized, and no conclusions can be gathered from it. Despite seeming promising at first, further research is needed before any substantiated claims can be made.

Paper B

Players actively explored all available dialogue options, revisited dialogue branches, and reported feeling able to shape the protagonist’s attitude and relationships. This supports referenced literature claim that even inconsequential choices (those that do not change the

final outcome) can meaningfully support a sense of control when they enable role-play and character interpretation.

Technical and UI friction, such as confusing interaction ranges, lack of feedback on mouse clicks, broken dialogue paths, looping conversations, and the game artefact crashing repeatedly interrupted play. These issues forced players to repeat content and made the underlying system visible, shifting the attention from developing personal narrative agency to inquiring about the technical stability of the game artefact and its systems, which broke immersion and the illusion of a responsive narrative.

The study concludes that perceived agency depends on the combination of narrative framing, UI clarity, and technical stability, instead of solely on the presence of multiple dialogue options available. Even a well-written dialogue structure will fail to sustain agency if feedback is unclear, or the system behaves unreliably.

The research paper's main contribution is the developed insight that perceived agency in dialogue-driven games relies on frictionless technical and UI delivery before it can be meaningfully subverted.

Paper C

The pilot test showed that players quickly understood the *Guilt-o-meter* and treated it as an important part of the game. However, they relied on it too much instead of making decisions based only on dialogue. This suggests the system influenced how players made judgments, even in a very simple version of the game. The short interaction also limited emotional connection, which led to weak character attachment.

In the main playtest with seven participants, results showed mixed emotional responses. Some players said the *Guilt-o-meter* strongly affected how they interpreted characters and made decisions, while others felt it only had a small impact. Emotional reactions to the NPCs also varied: some players felt guilt or discomfort when firing employees, while others felt little or no emotional response. A few participants said the characters felt real and meaningful, while others saw them more as gameplay elements.

Overall, the results show that the *Guilt-o-meter* influenced player judgment and moral decision-making, but emotional impact depended on how connected players felt to the characters and how immersed they were in the experience

Due to technical difficulties during development and time constraints within the production schedule, the planned second iterative playtest following feedback from Playtest 1 could not be completed. This limitation reduced the number of refinement cycles available for evaluating design changes across multiple iterations.

Key Results

Across all three papers, a shared result is that players stay engaged with narrative-driven systems, but their experience is shaped by different factors working together.

Paper A and Paper B both show that players often still feel a sense of agency even when choices do not change the final outcome. Players either role-play, experiment with options, or explore dialogue to understand the story better. This suggests that even limited or inconsequential choices can still feel meaningful to players if they are well presented.

Paper B and Paper C connect through the importance of system quality and emotional design. Paper B shows that technical issues and unclear UI can break immersion and reduce the feeling of control, even if the narrative design is strong. Paper C shows a similar pattern in emotional terms: the *Guilt-o-meter* affects decisions, but only when players are immersed and emotionally connected to the characters.

Paper A and Paper C also connect in showing variation in player behavior. Some players fully engage with the experience, while others treat it more as a system to test or observe. In Paper C this appears as differences in emotional response, where some players feel guilt and others remain detached.

The key connection between the papers is that player experience is not determined by narrative design alone. Instead, it depends on a combination of choice structure, technical stability, and emotional engagement. When these work together, players feel more agency, immersion, and moral involvement.

6 Discussion & Conclusion

Paper A focuses on how the approach to a medium greatly impacts interactions had with it. Exploring a player's connection to their in-game character is a way of exploring that approach as the character acts as a lens. According to existing literature (Ferchaud & Oliver, 2019; Lewis et al., 2008) players' behavior is impacted by the character they embody at the time. Finding a way to use these phenomena deliberately in a game design context could allow for emotional priming. This priming could be used in creative ways to create harmony or dissonance among other uses.

Each research question explores a part of a larger question that being the application of *Affective Narrative Application to Games Involving Narrative Choices*. Paper A focuses on approach and perspective, the lens through which the experience is processed and how people get immersed into space. Paper C focuses on leveraging immersion and investment to explore moral decision-making, reinforcing the experience by requiring effort and investment on the part of the player. Finally, paper B investigates the culmination and anticlimax when previously established patterns and promises explored in papers A and C are not delivered on. Each stage exploring different but linked approaches to designing for human emotion.

This study examined how narrative-driven choices and emotional feedback systems influence players' moral composure, particularly guilt and moral reflection. The findings suggest that the "Guilt-o-meter" significantly shaped how players interpreted moral decisions, demonstrating that interface systems can actively influence ethical reasoning during gameplay.

This aligns with Gibbons (n.d), who describes the evolution of moral choice systems from simple binary structures toward more refined designs that encourage interpretation and emotional engagement. In this study, the "Guilt-o-meter" functioned not as a neutral indicator

but as an interpretive tool that guided player judgment, reinforcing the idea that modern moral systems shape perception rather than simply present outcomes.

The results also support Liu (2025), who argues that moral choices in games can foster ethical reflection by making decisions feel personally meaningful. Participants who engaged more deeply with the narrative reported hesitation, discomfort, and responsibility, suggesting that guilt emerges when players emotionally invest in characters and situations.

However, the study also found that emotional feedback systems alone are insufficient. Some players reported limited emotional response due to short interactions and weak character development. This indicates that while systems like the “Guilt-o-meter” influence interpretation, meaningful moral reflection depends on narrative depth and immersion.

Interestingly, some participants trusted the “Guilt-o-meter” more than their own judgment, showing how players may treat emotional interfaces as authoritative. Within the broader view these findings highlight that emotional systems do not just support narrative interpretation but actively shape moral experience in real time

The primary finding is that narrative is highly sensitive to its delivery, as even minor issues in UI feedback, interaction flow, or technical stability can quickly diminish the player’s sense of agency and personal narrative. This aligns with Eriksson Turtiainen’s (2018) claim that inconsequential choices can support agency only as long as players experience them as meaningful, moment-to-moment opportunities to express their character. When players encounter unclear interaction prompts, looping dialogue, or system crashes, their attention shifts away from making choices and constructing a personal narrative toward examining the inner workings of the system itself. Kiander’s (2019) “beads on a string” model likewise assumes that each local segment feels mechanically coherent; in this study, looping conversations and technical failures made the underlying string visible too early, drawing the player’s attention away from choices and toward the exposed system structure. In these

moments, the illusion of choice, the player's personal narrative, and their perception of the player character often collapsed pre-emptively, before the intended narrative subversion could take place.

A faulty Windows operating systems update corrupted two research team members' development machines, leading to repeated system crashes and loss of development progress. Attempts to mitigate the damage done have taken a significant development allocated time.

Due to the lost time mitigating technical difficulties, the research team could not develop the game artefact to the intended scope and had to revert to improvised solutions and rushed research.

Due to various setbacks experienced by the research team, the rigid narrative resolution could not be empirically tested. The future research should examine how different forms of fixed endings affect perceived agency, frustration, and narrative satisfaction. The next step is to repeat the playtests with a stable build that features the rigid narrative resolution, while combining observation, interviews, and brief questionnaire to compare player responses before and after the resolution.

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7.1 Ludography

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8 Appendix (A)

8.1 Pilot Test Questionnaire

Q: What are your thoughts about the dialogue presented during the playtest?

Q: What are your thoughts about the dialogue choices presented during the playtest?

Q: What are your thoughts about the overall narrative presented during the playtest?

Q: Did the Guilt-o-Meter influence how you felt about the people you were talking to or did you form your own opinion through the conversations?

Q: Did the conversations with the NPCs make you feel anything towards them or did they feel like obstacles to get through?

Q: At any point during the conversations did you feel uncomfortable about what you were being asked to do even knowing it's just a game?

Q: What impression, if any, did you get of the character you play as?

Q: When speaking to the characters in the game did you make choices from the perspective of your player character or did you play as yourself?



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Examining Behavioral Differences Based on a Players Degree of Attachment to an In-Game Avatar.

Faculty of Arts
Department of Game Design
Author: Alison Payne
Bachelor Thesis in Game Design, 15 hp
Program: Bachelor's Programme in Game Design and Project Management - K5375
Supervisor: Josephine Baird
Examiner: Ray Whitcher
June, 2026

Abstract

This paper examines how the degree to which a player feels connected to a player's avatar impacts their behavior. The goal is to isolate specific correlations in order to allow for more intentional design regarding a player's perspective. This is done by observing participants interacting with a first-person pseudo detective mystery game and then followed by a semi-structured interview. Research through design is the primary methodology and a modified version of the Character Attachment (CA) framework as defined in Lewis et al. (2008) and the 7-factor model (Qin et al., 2009) will be used to aid analyzing the data. Findings were in line with existing literature. A larger sample size and a greater focus on qualitative data.

Character Attachment, Role Distance, Role Play, Research Through Design, Video Games

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1 Introduction

Games are an interactive medium that can take many forms such as physical in the form of board games and sports but also digital such as with video games. Each of these allow players to temporarily embody a character distinct from themselves. These characters can vary widely from a soldier in a modern first-person shooter game (FPS), to a doctor in a live action roleplaying game (LARP). Whether it is in digital or physical form this fictional setting temporarily changes the rules of the situation which suggests behavior may be impacted. Existing literature suggests that this embodiment or projection has a noticeable impact on a player's behavior (Ferchaud & Oliver, 2019; Lewis et al., 2008). However, existing research has not examined these behavioral differences as a design feature to the extent where they can be deliberately used to help design an experience. This is an important question because if players respond predictably to the degree of projection, designers could be more deliberate in their use of the player character to evoke specific emotions or lack thereof. Building on that this paper will examine the research question;

> *To what extent does a player's ability to project onto the in-game avatar impact their behaviour / interactions within digital games.*

To test this research question, and the research questions for the other papers in this KAPPA, the research team have created a first-person pseudo detective mystery game set in an office by the name of *Guilt-o-Meter*. Creating a game artefact was a necessary component of our use of Research Through Design (RTD) as our methodological approach. To analyze and interpret the data categories from the Character Attachment framework (CA) (Lewis et al., 2008), and the 7-factor model (Qin et al., 2009).

2 Background

The connection people have with fictional characters has long been a topic of interest. Whether the character is from a video game, a roleplaying session or even a book, some degree of connection is expected, however, it manifests in different ways. The connection a person may feel towards a character from a television series may manifest as a parasocial relationship, while a player may feel a sense of attachment to the avatar they just created. This connection varies in intensity and is based on the medium. In the context of video games the player often takes the perspective of an entity in the world, these entities, or characters, have varying degrees of preexisting character traits, ranging from blank slates that a player can design themselves such as in *Fallout 76* (Bethesda Game Studios, 2018), to fleshed out characters with desires and goals beyond the players such as Geralt from the *Witcher 3: Wild Hunt* (CD Projekt Red, 2015), this phenomena is discussed in Morality inside the matrix: A qualitative exploration of gamers' moral considerations in video games (Ferchaud et al., 2022). Previous research has suggested a link between a player's ability to connect/attach with a character and enjoyment (They May Be Pixels, But They're MY Pixels:" Developing a Metric of Character Attachment in Role-Playing Video Games [Lewis et al.], 2008), as well as a tendency for players to disassociate themselves from a character when forced to make "Bad" or "Evil" choices (Ferchaud & Oliver, 2019). This suggests a noticeable connection between players and their point of view (POV) characters.

Based on that, this research aims to isolate and measure this connection in a first-person office detective mystery game artifact specifically created for this purpose. The game artifact has a minimalist style and requires players to speak to a couple of "non-playable characters" (NPCs) while trying to discover which of the two employees is responsible for stealing money

from the company. When the player is satisfied they have found the culprit they choose who to fire, however, there are twists and information that make this decision more difficult. Using games such as *Papers, Please* (3909 LLC, 2013) as inspiration the research team wanted to make a game that would force players to make difficult and uncomfortable choices. The research team chose to make a pseudo detective mystery game for a of couple reasons: primary among them being the requirement for players to engage with the narrative and introducing an unreliable "narrator" in the guilt-o-meter to further complicate decision making.

This is important because if a player's degree of attachment to a character has a noticeable and predictable impact on their behavior it could open a new field of study for game design. With a greater understanding of the concept more designers could use the players' connection to character with the same intentionality used for other design considerations such as color or level design.

3 Research Design

To test the research question, the research team focused on self-expression and immersion. Since both Character Attachment (CA) as outlined in Lewis et al., (2008), and the 7-factor model (Measuring Player Immersion in the Computer Game Narrative [Qin et al.], 2009), have some form of expression of player agency, both focusing on control, therefore designing with agency and expression in mind is a viable approach. Immersion is crucial for (CA) as it (CA) is itself a form of immersion. A certain degree of immersion is a prerequisite for all 5 of the criteria outlined in Lewis et al. (2008). To test for expression, I wrote dialogue options with varying degrees of personality, there would always be standard responses or questions but there would also be additional options ranging from humorous to intense or just eccentric. These options allow players to feel like they are deviating from the default path with intent of

providing a sense of autonomy and self-expression. They can simultaneously act as a catalyst to express the personality of the player character effectively showing the range of what this character would say in a given situation.

3.1 Tools and Techniques

For data collection the research team used an 8 question semi-structured interview, supervised playtests (to observe players' behavior during gameplay), consent forms, and any audio recording device (for recording the playtest and subsequent interview). Convenience sampling was the method to gather participants for our research.

3.2 Methodology and Theory

The research team utilized Research Through Design (RTD) as the primary methodology followed by data gathering with an 8 question semi-structured interview, and observations gathered during the playtests. Research Through Design (RTD) is a methodology that focuses on iteration where the development itself creates data.

The primary method of data collection was an 8 question semi-structured interview. We conduct semi-structured interviews with the intent for more flexibility with follow-up questions while still retaining a fallback script. Additionally, the semi-structure format allowed each researcher to have their questions answered while allowing other researchers to have enough flexibility to ask follow-up questions as necessary.

During the playtests participants were asked to vocalize any thoughts they had, while a researcher was present to observe and document the gameplay and any comments participants expressed.

To analyze the gathered data the Character Attachment (CA) framework (Lewis et al., 2008) defined as;

"we define CA as an individual's feelings of (a) friendship and (b) identification with a video game character when an individual (c) is willing to suspend disbelief, (d) feels responsible for the game character, and (e) feels in control of the game character's actions." (Lewis et al., 2008, p. 516).

(CA) was used to help contextualize and attempt to analyze the data gathered.

To measure immersion, inspiration was taken from the 7-factor model as defined in (Qin et al., 2009). This research aims to look for signs of curiosity, comprehension, challenges / skills, empathy concentration, control and familiarity. Due to the scope of this study certain data will not be gathered. Due to this the model will not be used in its entirety.

3.3 Limitations

The limitations of this study are the following; scope, limited data collection and experimental validity.

Scope is a broad category that covers the number of participants, the number of questions and the scope of the game. Considering the limited time and resources, we were unable to conduct more playtests, while a larger dataset is not strictly necessary it would help with determining statistical significance and experimental validity overall. Similarly, the limited number of questions (8) where only 2 were specifically pertaining to this research question made it difficult to interpret the data according to the Character Attachment (CA) framework (Lewis et al., 2008).

Due to the lack of a control group and data being entirely self-reported, it is difficult to determine if the data is generalizable, which is furthered by the use of convenience sampling primarily targeting other students studying game design.

Finally, due to the RTD framework and the number of iterations and variation data between the different versions of the game will vary and compensating for these differences was not a core focus of the data analysis procedure.

3.4 Author Positionality

Due to the use of convenience sampling, researchers often had pre-existing personal relationships with the playtesters, which risked skewing or altering the data. Due to one researcher being responsible for much of the front-end development of the game artifact playtesters aware of this fact may be hesitant to criticize elements such as the writing or visual presentation in that researcher's presence.

The playtests took place in neutral public locations such as booked rooms at libraries; this was to minimize external variables as well as to avoid using spaces that could cause power imbalances or make participants uncomfortable.

The researcher responsible for writing for the game artifact had pre-existing expectations as to how players would or should feel in response to various sections of dialogue meaning my observations may be impacted. To mitigate this 2/3 of all playtests were conducted by other researchers so data could be cross-referenced and compared.

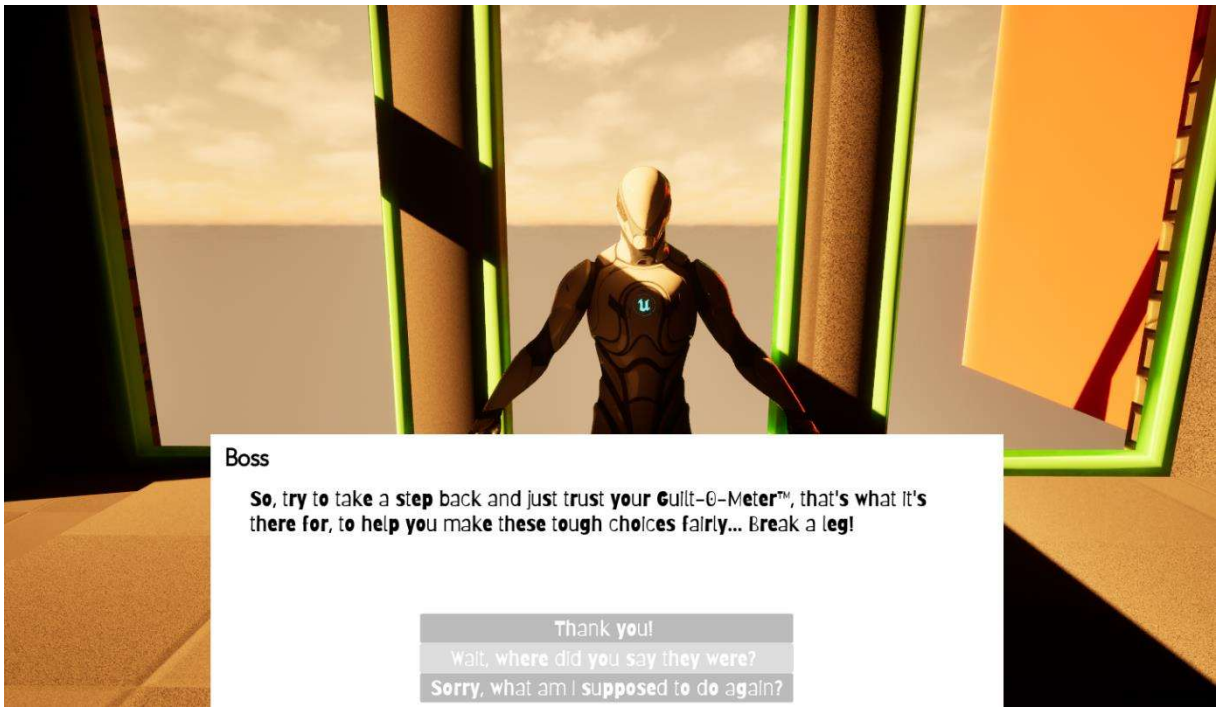
4 Results & Discussion

4.1 Pilot Test

The pilot test, while limited, still showed promise. Despite only aiming to prove the dialogue mechanic, participants were engaged in what narrative was present and some expressed interest in playing later versions of game. One of the most important factors for my research is immersion as it is difficult to inhabit a character if their context is not immersive to some degree.

Figure 1

State of the Dialogue System During the Pilot Test.



Based on the results from the pilot test proving the dialogue mechanic as well as the narrative we felt confident in moving forward with development. This meant adding additional characters, polishing, texturing and finishing the first level for iteration 1. The primary focus was finishing the characters' dialogue trees and making the space feel more appropriate to aid with the storytelling. This was to increase immersion while also expanding on the features participants expressed interest in Qin et al. (2009).

One major design decision was to portray the characters in a very minimalistic way, while this was in line with one of our stylistic inspirations *Jazzpunk* (Necrophone Games, 2014), it also meant we could use blank faces forcing players to interpret a character's emotions based on dialogue or other external factors, see Figures 2,3. This decision also saved a lot of time on texturing allowing for a greater focus on research critical elements such as writing.

Figure 2

Character "Amanda" in the Breakroom, Iteration 1



Figure 3

Character "Carlos" in the Office, Iteration 1



Another takeaway from the pilot test was needing to make the space feel appropriate as playtesters expressed confusion as to where they were supposed to be, one briefly believing it to be a spaceship. This led the research team to prioritize environmental art more than what was initially planned. This also allowed us to tell the story through the environment rewarding players who explore with extra clues to solve the mystery while reinforcing the detective mystery theme.

Figure 4

Boss's Office (Before), Post Pilot Test



Figure 5

Boss's Office (After), Iteration 1



4.2 Iteration 1

Iteration 1 was a massive leap from the pilot test, with a nearly complete, fully textured and decorated level players had a much easier time understanding and getting immersed in their surroundings. Notably, every playtester had a distinct impression of the character they were playing, and some actively made decisions from the perspective of that character. This, along with comments about the setting and narrative, suggests participants were at least somewhat

immersed in the game. This is further reinforced by looking at how the answers map onto the categories of the 7-factor model (Qin et al., 2009).

Figure 6

Table showing the number of participants (n/4) who expressed experiencing each category of the 7-factor model.

Factors	n/4 Participants Expressing Factor
Curiosity	2
Comprehension	4
Challenge / Skills	N/A
Empathy	3
Concentration	4
Control	4
Familiarity	N/A

Additionally, 3 participants explicitly used the term “engaging” to describe the playtest, and one used the term “immersive”, which does suggest that participants were sufficiently immersed to have a meaningful connection to the player's character.

The results line up with the expected results, being that players bring their own moral frameworks into the game (Ferchaud et al., 2022). This included meta-gaming as described in (Ferchaud et al., 2022).

The results from the playtests in iteration 1 were very positive. Based on the responses, it seems like we succeeded, however, there was still room for improvement. Many playtesters mentioned bugs or mistakes that negatively impacted the game. Additionally, only one of the two levels was implemented, and a proper ending was missing. On the technical side the research team had yet to implement a memory system to the dialogue, meaning each time a

player spoke to an NPC they would be treated like it was the first time they met, which impacted immersion for some participants. The research team managed to meet almost all of those goals in time for iteration 2, unfortunately, the research team suffered major technical difficulties preventing implementation. Due to this we never conducted any iteration 2 playtests, and the effect of all of the improvements is purely speculative.

5 Discussion & Conclusion

While the data gathered is in line with previous literature, it is important to consider the experimental validity of these tests. Firstly, the number of participants is very low with only 6 total and 4 documented participants for iteration 1, and a sample size of 3 for the pilot test, it is not possible to know if the results are generalizable. This is further complicated due to the vast differences between the pilot test and iteration 1, meaning comparing the two data sets to each other is not particularly useful. Additionally, the use of convenience sampling limits the generalizability even further.

Secondly, both CA and 7- factors utilized specialized surveys to gather and standardize data, creating or using one of the existing surveys would aid greatly in standardizing the data and making it more generalizable and comparable to existing research. A quick survey post playtest with specific questions would have been helpful in interpreting playtester feedback. In short, supplementing qualitative data with quantitative data would allow for a more complete picture overall.

Considering the research question, having different conditions as well as a control group would be ideal. Being able to compare a control group to a third person version or a version with a character creator would be perfect for pinpointing specific differences which would allow for even more deliberate use of perspective in a design context.

The lack of an iteration 2 meant large portions of the game were absent during every playtest conducted.

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6.1 Ludography

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7 Appendix (A)

7.1 Research Questionnaire

Q: What are your thoughts about the dialogue presented during the playtest?

Q: What are your thoughts about the dialogue choices presented during the playtest?

Q: What are your thoughts about the overall narrative presented during the playtest?

Q: Did the Guilt-o-Meter influence how you felt about the people you were talking to or did you form your own opinion through the conversations?

Q: Did the conversations with the NPCs make you feel anything towards them or did they feel like obstacles to get through?

Q: At any point during the conversations did you feel uncomfortable about what you were being asked to do even knowing it's just a game?

Q: What impression, if any, did you get of the character you play as?

Q: When speaking to the characters in the game did you make choices from the perspective of your player character or did you play as yourself?



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The Impact of Rigid Narrative Resolution on Perceived Player Agency

Establishing Narrative Agency Through Dialogue Mechanics

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Abstract

This paper investigates how the "illusion of choice” in video games is created through dialogue mechanics and then disrupted by a predetermined rigid narrative resolution in a story-driven game. Utilizing the Research through Design (RtD) framework, the study focuses on the technical delivery of narrative through User Interface (UI) implementation and programming. The research evaluates, through playtesting, the game artifact *Guilt-o-meter* and examines how players interact with the game’s branching dialogue system that’s designed to build agency, followed by a singular rigid narrative resolution that purposefully breaks it.

Findings from initial playtesting of *Guilt-o-meter* reveal that responsive dialogue mechanics and diverse options successfully generate high initial player agency and immersion. However, UI clarity and technical issues significantly impact the mechanical delivery of the narrative and can shatter the illusion of choice preemptively.

By applying Tula Eriksson Turtiainen’s (2018) framework of “meaningful” and “inconsequential” choices and Verna Kiander’s “beads on a string” model (2019), this paper establishes a theoretical baseline and design framework to measure the psychological impact of subverting player agency in future studies.

Keywords: Branching Dialogue, Illusion of Choice, Inconsequential Choices, Player Agency.

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Introduction

A critical challenge in narrative-driven game design is balancing the player's desire for agency with the developer's need for a structured storyline. This paper investigates the research question: How does a rigid narrative resolution undermine the perceived agency established by branching dialogue mechanics in *Guilt-o-meter*?

While the original scope aimed to test a fully realized final rigid narrative resolution, unforeseen critical development setbacks forced the research team to pivot the game artifact's development. A placeholder "rigid narrative resolution" was quickly implemented, but due to the team running out of time for research, this temporary rigid narrative resolution was not actually playtested. Because of this, this paper evaluates the successful establishment of narrative agency during the initial playtests and details the technical framework designed for the untested final "rigid narrative resolution".

The theoretical framework draws on Tula Eriksson Turtiainen's (2018) distinction between "meaningful" and "inconsequential" choices, Barisbi Alborov's analysis of the illusion of choice, and Verna Kiander's (2019) "beads on a string" structural model. The methodology utilizes a Research through Design approach, generating insights from qualitative data gathered during the first iteration of playtesting.

1 Background

The game artifact *Guilt-o-meter* builds on the "bureaucracy simulator" genre in which the players take on administrative roles within restrictive institutional systems. In these games, players typically navigate moral and procedural dilemmas from a position of limited power, making decisions under systemic and organizational constraints. Well-known titles in this space, such as *Papers, Please* (Pope, 2013), *Beholder*, and *Not Tonight*, place players in

morally challenging bureaucratic roles where choices must be made under constant pressure, *Guilt-o-meter* draws on this tradition while focusing specifically on a single rigid narrative resolution. Many existing games in this genre offer multiple narrative resolutions to reward player choice and emphasize long-term narrative consequences. In contrast, this project deliberately explores the design space of narrative inevitability: a structure in which the player is encouraged to feel in control during moment-to-moment interactions, while the overarching narrative resolution remains fixed and agnostic to player agency.

To evaluate how this structure delivers its narrative, the study draws on three theoretical frameworks. The first is Tula Eriksson Turtiainen's (2018) distinction between meaningful and inconsequential choices. Eriksson Turtiainen (2018) examines how different types of story choices affect a player's sense of control, arguing that "meaningful" choices are those that tangibly alter the game state, while "inconsequential" choices leave the objective outcome unchanged, but still allow the player to role-play and reflect on their character's motives. Inconsequential choices can therefore support agency if they shape how the player understands their avatar and relationships, even when they do not branch the plot.

The second framework is Alborov's analysis of the illusion of choice. Alborov (2017) argues that a carefully constructed illusion of choice can be preferable to fully branching narratives, because it provides a psychological feeling of agency without the unsustainable production costs of maintaining many divergent storylines. However, Alborov (2017) also warns of the fragility of this strategy, as once players realize that their decisions have no meaningful consequences, the illusion breaks, and they may experience disappointment or a sense of betrayal. This tension between efficient authorial control and the risk of exposing the illusion is central to *Guilt-o-meter* design.

The third framework is Kiander's (2019) "beads on a string" structure. In this model, local segments of play (the beads) may vary according to the player's choice, but the overarching narrative string remains fixed, drawing all paths back to a shared outcome. The structure supports replayability and local agency while preserving the overall story. Kiander (2019) also notes that the effectiveness of this structure depends on players believing that their choices matter within each "bead", and if they perceive that every branch collapses back into the same outcome, the illusion of a responsive narrative can weaken.

Guilt-o-meter uses these frameworks by leaning heavily on inconsequential choices embedded within a fixed narrative string. Its branching dialogue system is designed to give players opportunities to express the protagonist's attitude, experiment with different responses, and explore the morality of a bureaucratic workplace, even though the rigid narrative resolution is predetermined and fixed. This background motivates the central question of the thesis: how a rigid narrative resolution might undermine the perceived agency established through the branching dialogue mechanic.

2 Research Design

This study employs a **Research through Design (RtD)** framework, using iterative development of *Guilt-o-meter* to generate knowledge about the relationship between branching dialogue, rigid narrative resolution, and perceived player agency. Within this framework, design, implementation, and evaluation are treated as interlinked activities, where changes to the game artifact directly inform the research questions and vice versa.

Data for the first iteration were collected through a combination of formal playtests, semi-structured interviews, and observational notes taken during and immediately after the play sessions. Playtest 1 consisted of five volunteer participants of various age groups and

fields of study and occupation, each playing the presented vertical slice for approximately 10 minutes. During the sessions, participants were encouraged to “play as they normally would” while thinking aloud when possible, allowing the research team to capture moments when players felt in control or were confused by the dialogue system.

For the first round of testing, the game artifact functioned as a vertical slice that showcased the dialogue system rather than the full narrative arc. The prototype utilized Behavior Trees, Unreal Engine’s node-based system for controlling non-player character (NPC) decision-making and sequencing actions, to govern NPC dialogue flow. The prototype utilized Behavior Trees to govern NPC dialogue and Dialogue Data Assets to populate and manage branching conversations, allowing for multiple dialogue options and conversational paths to be presented without a fully implemented rigid narrative resolution or game state-altering decision-making. Each Behavior Tree dynamically created a dialogue UI widget at runtime as part of active NPC behavior, pulling the appropriate lines from a Dialogue Data Asset attached to the said NPC, which contained multiple dialogue options and allowed for conversational paths to be presented.

Because the rigid narrative resolution was not yet implemented, the focus of Playtest 1 was on how effectively the user interface and branching dialogue mechanics established perceived agency during moment-to-moment interactions.

Before each playtest and interview, participants received an information sheet explaining the purpose of the study, the play session, and the follow-up interview. All participants provided informed consent to participate in the study. Participants were informed that participation was voluntary and that they could stop at any time without penalty. All participants are anonymized in this thesis, and recordings and notes were stored on a

password-protected device accessible only to the research team and will be deleted after the completion of the degree project.

After each play session, participants completed a semi-structured interview based on a shared questionnaire, covering topics such as perceived freedom of choice, clarity of feedback, and moments of satisfaction or frustration. The participants' answers were recorded, and in-session notes were taken documenting notable behaviors such as repeatedly cycling through all dialogue options or pausing when interaction prompts were unclear. The qualitative material was then reviewed collaboratively by the research team and grouped into recurring themes related to perceived agency, UI clarity, and technical stability.

The research and development cycle faced unforeseen technical and logistical limitations that affected both the game artifact and the data that could be collected. Following Playtest 1, the research team member responsible for writing dialogue design space of narrative inevitability and visual assets creation experienced a medical emergency, which stopped narrative production. Shortly afterwards, two other team members suffered critical hardware failures that corrupted their computers. To mitigate the effect of this, one team member migrated to a different operating system, but in the final week of development, cross-platform compilation issues emerged that prevented other team members from accessing the updated project files.

These setbacks forced an improvised implementation of a placeholder with rigid narrative resolution in which a seemingly consequential firing decision funnels into a fixed outcome. However, due to the compounded delays, there was no remaining time for the research team to conduct a second round of playtesting with this new iteration. Consequently, this study's empirical data is limited to Playtest 1 and focus on the establishment of perceived agency rather than its later subversion. The implications of the rigid narrative resolution are

therefore discussed primarily on a theoretical basis, informed by Alborov's (2017), Eriksson Turtiainen's (2018), and Kiander's (2019) frameworks, rather than on direct empirical measurement of player frustration.

3 Results

While the subversion of player narrative agency through the rigid narrative resolution could not be empirically tested, the iterative testing and design phases of Playtest 1 provided detailed insights into the construction and reception of the dialogue mechanic. The results below are organized by iteration, focusing on how players engaged with the branching dialogue, how they understood their agency, and which technical issues constrained that experience.

1. Iteration 1

The programmed dialogue branches in the vertical slice successfully established a strong initial sense of agency. Participants consistently attempted to exhaust all available dialogue options with each NPC, returning to previous nodes to explore different responses and see how characters would react. Observational notes and interviews describe players quickly becoming immersed in conversations and commenting positively on the range and tone of the options presented to them. During post-session interviews, participants reported that the dialogue choices made them feel that they could shape the protagonist's attitude and relationships, even though they did not yet see long-term narrative consequences.

Despite the high engagement, several mechanical and interface issues disrupted the illusion of choice during moment-to-moment play. Participants frequently expressed confusion about when they were within the interaction range of an NPC or clickable object, often clicking the mouse repeatedly without visible feedback. One playtester noted that "clicking doesn't seem to do anything... would be nice if there were an indicator", pointing directly to a lack of clear affordances around dialogue initiation and progression. In multiple

sessions, players hesitated or circled around NPCs while searching for the correct interaction spot, which pulled attention away from the content of the dialogue and narrative.

Additional technical problems further weakened the continuity of perceived agency. Participants encountered broken dialogue paths featuring placeholder lines, options that unexpectedly reset conversations to their starting state, and one severe issue where talking to a specific NPC after completing its dialogue caused the game artifact to stop functioning entirely. These interruptions often forced players to repeat sections of dialogue they had already explored, producing frustration and highlighting the underlying structure of the system rather than the illusion of a responsive narrative. As a result, the same branching system that initially supported agency also exposed its own structure when technical stability was compromised.

2. Iteration 2

To address the development setbacks and move towards the original research aim, the research team implemented a placeholder rigid narrative resolution for Iteration 2. In this version, the player's accumulated agency through branching dialogue was designed to culminate in a morally loaded choice of which NPC to fire. Immediately after the player confirmed this "meaningful" decision, the game would present the fixed statement "You will always be an intern" followed by a single "Quit game" option, effectively collapsing all potential narrative paths to a single outcome.

Due to the time lost to medical and technical issues, this iteration was not playtested, and no observational or interview data were collected about player responses to the rigid narrative resolution. Consequently, this section can only describe the intended behavior of the rigid narrative resolution rather than reporting empirical findings. The potential impact of this

design on player frustration and perceived agency is therefore discussed on a theoretical basis in the Discussion & Conclusion section, rather than in the Results section.

Discussion & Conclusion

Although this study could not fully answer the research question in its originally intended form, Playtest 1 still generated useful design insights about how perceived agency is established and how quickly it can be undermined. The findings suggest that branching dialogue is not sufficient to sustain a sense of narrative agency. Rather, perceived agency depends on the combined effect of narrative framing, interface clarity, and technical stability. When these supporting systems function effectively, players can experience even inconsequential dialogue choices as meaningful in the moment because they help define the player character's attitude and relationship to the game world.

This finding aligns with Eriksson Turtiainen's (2018) distinction between meaningful and inconsequential choices. Eriksson Turtiainen (2018) argues that even choices that do not alter the final game state can still support a player's sense of control when they enable role-play and shape how the player understands their character. The results of Playtest 1 support this claim: participants explored multiple dialogue options, engaged closely with NPC interactions, and described the conversations as giving them influence over the tone of their interactions. In this sense, the dialogue system succeeded in establishing a perceived agency even before any larger narrative consequences were introduced.

The playtest results also show that this perceived agency was fragile. Confusion around interaction ranges, lack of clear UI feedback, broken dialogue paths, and critical bugs repeatedly interrupted the player's experience and redirected attention from narrative choice to system failure. This is significant because it suggests that the illusion of agency is not

broken only by narrative outcomes, but also by technical friction in the delivery of those outcomes. This means that branching dialogue does not create agency by itself, but that it must be supported by readable interface cues and dependable implementation to maintain the player's belief that the game is responding meaningfully to their input.

This can also be understood through Kiander's "beads on a string" model. Kiander (2019) describes a structure in which local choices may vary while the larger narrative returns to a stable path, allowing the player to experience short-term agency without requiring fully divergent storylines. The results of Playtest 1 suggest that this model can work only when the "beads" feel mechanically coherent. When dialogue loops unexpectedly, placeholder lines appear, or progress breaks entirely, the player becomes aware not only of a guided narrative structure, but also of the artificial system beneath it. In that situation, the narrative string becomes visible too early, and immersion is weakened before the larger structure can achieve its intended outcome.

Iteration 2 was designed to push this tension further by confronting the player with a morally weighted firing decision and then collapsing every path into the same singular rigid narrative resolution. This design decision directly engages with Alborov's warning that the illusion of choice can create disappointment once players realize their choice has no meaningful consequence. It is also informed by Tula Eriksson Turtiainen's (2018) finding that players can react negatively when they are forced onto a path after believing they had direct control. However, because Iteration 2 was not empirically tested, this paper cannot claim that players experienced frustration, betrayal, or diminished agency in response to that rigid narrative resolution.

The main contribution of this study is not a conclusive measurement of how rigid narrative resolution undermines player agency, but a design insight: perceived narrative

agency in dialogue-driven games relies on frictionless technical delivery before it can be meaningfully subverted. The study shows that even well-structured branching dialogue and role-play-oriented choices are vulnerable to UI ambiguity and system instability, which can prematurely break the player's sense of control before any intended narrative payoff occurs. This suggests that subverting agency is only effective after agency has first been convincingly established through both narrative and technical design.

This study features clear limitations, with the most significant being that the rigid narrative resolution central to the research question was implemented only as a placeholder and was not playtested. In addition, the qualitative findings are drawn from the first iteration of the game artifact only, meaning the conclusions are based on a limited sample of player interactions with a vertical slice and not a complete game artifact. As a result, this paper should be understood as an exploratory practice-based study that identifies a promising research direction rather than definitively resolving the psychological impact of rigid narrative resolution.

Future work should focus on directly testing the rigid narrative resolution with a second playtest round using a stable build and clearer UI feedback. Additionally, the next step would be to compare player responses before and after the rigid narrative resolution using interviews, observational data, and a questionnaire regarding perceived control, sense of frustration, and narrative satisfaction. This would allow the theoretical claims drawn from Alborov (2017), Kiander (2018), and Eriksson Turtiainen (2019) to be examined more closely in relation to the specific design of *Guilt-o-meter*.

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5 Appendix (A)

The following semi-structured interview questions were used after each play session in Playtest 1:

1. What are your thoughts about the dialogue presented during the playtest?

2. What are your thoughts about the dialogue choices presented during the playtest?

3. What are your thoughts about the overall narrative presented during the playtest?

4. Did the Guilt-o-Meter influence how you felt about the people you were talking to or did you form your own opinion through the conversations?

5. Did the conversations with the NPCs make you feel anything towards them — or did they feel like obstacles to get through?

6. At any point during the conversations did you feel uncomfortable about what you were being asked to do even knowing it's just a game?

7. What impression, if any, did you get of the character you play as?

8. When speaking to the characters in the game did you make choices from the perspective of your player character or did you play as yourself?



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Moral Agency in Narrative Games

Exploring Moral Decision Making in Interactive Narratives

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Abstract

This study explores how narrative-driven choices in games influence players' moral composure, with a focus on feelings of guilt and moral reflection. It examines why such emotional responses matter by highlighting their relevance to both players and designers interested in creating more immersive and meaningful interactive experiences. Using a qualitative approach, data is collected through player interviews and gameplay recordings to capture both real-time decision-making and post-play reflections. The analysis compares the choices players make with their reported emotional responses, identifying patterns in when players experience guilt versus emotional detachment. Central to this research is the development of the "Guilt-o-meter", a visual game mechanic designed to measure and reflect the intensity of players' moral decisions. The findings are expected to show that players experience stronger guilt when choices feel personal and when narrative immersion is high. Additionally, the "Guilt-o-meter" may enhance players' awareness of consequences and deepen emotional engagement. The study concludes that emotional feedback systems can greatly enhance player immersion and encourage moral reflection, adding a quantifiable emotional component to affective narrative research in games and enhancing previous research on player viewpoint and the illusion of choice.

Keywords: affective game design, emotional engagement, guilt, interactive narratives, moral decision-making, player experience,

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1 Introduction

Narrative-driven video games increasingly rely on player choice to create meaningful and emotionally engaging experiences. This study investigates how such choices influence a player's moral composure, with a particular focus on feelings of guilt and moral reflection. As games become more immersive, understanding how and why players emotionally respond to their decisions is important not only for academic research but also for designers aiming to create impactful narratives. The central research questions ask how narrative-driven choices affect players' sense of moral responsibility, and to what extent players perceive these decisions as personal rather than as part of a scripted system.

The study is grounded in the theoretical frameworks of affective game design and moral decision-making in interactive narratives, which explore how emotions are elicited and shaped through gameplay. To examine these dynamics, a qualitative methodology is employed, including player interviews and gameplay recordings. A sample of participants is used to capture in-depth insights into players' reasoning and emotional reactions. Additionally, this research introduces the "Guilt-o-meter" a visual feedback system designed to measure and reflect the intensity of moral choices within gameplay.

This research contributes to a broader team project focused on affective narrative applications in games involving narrative choices. While other papers within the KAPPA project examine topics such as player perspective, narrative immersion, and the illusion of choice in interactive storytelling, this study specifically focuses on the emotional consequences of player decision-making, particularly guilt and moral reflection. The research is guided by the question: "How does moral agency in narrative games influence player decision-making and emotional responses, particularly feelings of guilt and moral reflection, within interactive narratives?". Together, the combined studies aim to explore how narrative systems, player agency, and emotional feedback mechanics influence player experience and engagement within interactive games.

2 Background

The use of moral choice systems in video games has evolved into one of the most influential design features in narrative-driven interactive media. Early moral systems in video games often relied on binary "good versus evil" frameworks with clear consequences for player actions, making decisions relatively predictable and limiting emotional ambiguity. *Gibbons, L. (n.d.). A history of moral choice in games* notes that early morality systems are commonly rewarded clearly with heroic or villainous behavior, while *Liu, Z. & The University of New South Wales, Sydney, Australia. (2025). The impact of moral choices in games on players' moral values* argues that more complex moral choices encourage deeper ethical reflection and can influence players' moral reasoning. Together, these studies highlight the shift from simple morality systems toward more emotionally complex interactive narratives. Over time, game design has shifted toward more complex moral frameworks that aim to reflect real-world ethical uncertainty and encourage deeper reflection. As discussed by Liam Gibbons (n.d) in his historical overview of moral choice in games, this evolution demonstrates a growing industry emphasis on player agency, where decisions are not only about outcomes but also about personal interpretation and emotional weight.

Alongside industry development, academic research has increasingly explored how moral choice systems affect players beyond gameplay mechanics. Z. Liu (2025) argues that moral decisions in games can influence players' moral values by encouraging reflection on ethical behavior within simulated environments. This suggests that games can function as spaces for moral experimentation, where players confront dilemmas that mirror real-life ethical reasoning. Affective game design further emphasizes the use of emotions such as guilt, empathy, and regret to deepen player immersion and strengthen narrative engagement, positioning games as both entertainment experiences and emotional systems.

Despite this, a gap remains in understanding immediate emotional responses after moral decisions. Much research focuses on narrative structure, branching systems, or long-term behavioral influence, rather than short-term emotional consequences. Feelings such as guilt or detachment are often underexplored, even though they strongly shape how meaningful a decision feels. Additionally, limited work explores how these emotions can be made visible within gameplay.

This study addresses these gaps by focusing on emotional outcomes of moral decision-making, specifically guilt and moral reflection. It introduces the “Guilt-o-meter”, a proposed mechanic designed to visualize emotional responses linked to player choices. Using qualitative methods such as interviews and gameplay recordings, the research examines how players interpret decisions and how these align with emotional reactions, identifying when guilt or detachment occurs in relation to immersion and perceived responsibility.

The theoretical foundation is grounded in affective game design and moral decision-making frameworks, explaining how interactive narratives shape emotional engagement. As Liu (2025) argues, “Video games are not just played, they are lived—players experience decisions as if they are their own, which makes ethical reflection more personal and impactful.” Therefore, guilt is expected to increase when choices feel personal and immersive, while detachment occurs when decisions feel system driven. The “Guilt-o-meter” may also enhance emotional awareness and reflection during gameplay.

This research contributes to a broader project by focusing on emotional outcomes, complementing studies on player perspective and the illusion of choice. While those examine interpretation and narrative structure, this paper adds an emotional dimension by focusing on post-decision feelings. The main limitation is the sample size and focus on selected games, which may limit generalizability. Despite the findings aim to inform how emotional feedback systems can enrich moral engagement and support more affectively aware game design.

3 Research Design

This study used a qualitative research design to explore how narrative-driven choices influence players' moral composure and feelings of guilt in interactive games. The research was conducted through the development and testing of a prototype 3D first-person office investigation game, inspired by *Papers, Please* (Pope, 2013), in which the player takes on the role of a floor manager tasked with identifying and removing employees "inconvenient" to the company." To assist in these decisions, players were provided with an in-game system called the "Guilt-o-meter", a visual user interface mechanic designed to react dynamically to dialogue choices and indicate the perceived intensity of guilt connected to each interaction.

The design process was informed by theories of affective game design and moral decision-making in interactive narratives. The game aimed to create emotional tension by placing players in morally uncomfortable situations where they had to judge and fire employees based on incomplete and falsified information. The "Guilt-o-meter" was developed as part of this research to visualize emotional feedback and encourage reflection on the player's actions. This mechanic functioned as both a gameplay system and a research tool, allowing the study to examine whether visible emotional feedback influenced players' sense of responsibility and guilt during decision-making.

Data collection consisted of one pilot testing session followed by one formal playtesting round. A total of seven participants took part in the playtest. Participants played the game for approximately five to ten minutes while gameplay sessions were recorded for later analysis. Following gameplay, semi-structured interviews were conducted to capture participants' emotional reactions, interpretations of the narrative choices, and reflections on the firing decisions within the game. The interviews focused particularly on whether players felt guilt, emotional detachment, or moral responsibility when removing employees and how the "Guilt-o-meter" affected their decision-making process.

The collected data was analysed through qualitative thematic analysis, comparing player actions during gameplay with their verbal reflections from the interviews. Attention was given to moments where players hesitated, questioned their decisions, or expressed discomfort after firing characters. The study also examined whether the "Guilt-o-meter" increased emotional awareness or encouraged players to think more critically about the consequences of their choices.

This work connects to the broader KAPPA project on affective narrative applications in games involving narrative choices. While the collaborative KAPPA chapter synthesizes the broader research themes, this paper specifically contributes an emotional perspective by focusing on guilt and moral reflection in gameplay.

Several limitations affected the study. Due to technical difficulties during development, the game could not be fully completed, and a planned second round of playtesting based on feedback from the first session could not be conducted. Additionally, the project was constrained by limited development time and tight course deadlines. The sample size also limits the generalizability of the findings. As the developer of the “Guilt-o-meter” and an active player of narrative games, the author acknowledges a personal interest in how games emotionally affect players, which may have influenced both the design process and interpretation of results.

4 Results

This chapter presents the findings from the development and playtesting of the prototype game, focusing on how players responded emotionally to narrative-driven moral decisions and the implementation of the “Guilt-o-meter” system. The results are structured around the main stages of development, including the pilot test, iteration process, and first formal playtesting phase. Attention is given to player immersion, emotional engagement, and the extent to which the “Guilt-o-meter” influenced player judgment and moral reflection.

4.1 Initial Design and Concept Development

The prototype was developed as a 3D first-person office investigation game in which the player assumes the role of a floor manager tasked with identifying and removing employees considered “inconvenient” to the company. The game’s central narrative involved a workplace investigation surrounding suspected embezzlement within the office. Throughout the game, the player interacts with employees, gathers information through dialogue, and ultimately decides who should be fired.

The original concept emerged from a design theme centred around trust and manipulation. During early brainstorming, the team explored how authority systems influence player judgment, which eventually evolved into a corporate investigation setting. The idea of the “Guilt-o-meter” was introduced as a company-provided tool intended to assist the player in identifying suspicious employees. However, rather than functioning as a purely objective truth detector, the system was designed to create uncertainty by subtly influencing the player’s interpretation of conversations and character behaviour.

The game was designed from a first-person perspective primarily for practical and immersive reasons. From a development perspective, first-person gameplay reduced the need to create and animate a player character model, allowing more development time to be focused on dialogue systems and environmental interaction. At the same time, the perspective increased immersion by placing the player directly within the office environment and making conversations feel more personal and immediate.

The gameplay loop involved walking through the office environment, interacting with employees through dialogue trees, observing the reactions of the “Guilt-o-meter”, and making decisions regarding which employee should be removed from the company. The “Guilt-o-

meter" reacted dynamically during conversations by visually changing in response to dialogue choices and perceived suspicious behaviour. The intention behind this mechanic was to examine whether emotional feedback systems could alter player perception and increase feelings of moral responsibility.

4.2 Pilot Testing

The pilot test focused on evaluating the dialogue system and the core concept behind the "Guilt-o-meter." At this stage, the game included only non-playable character (NPC) and a short dialogue interaction designed to test the emotional feedback mechanic.

The test showed that players immediately recognized the "Guilt-o-meter" as an important gameplay feature but also relied heavily on it rather than forming independent judgments through dialogue. One participant noted that their decisions still felt meaningful despite the prototype's simplicity, suggesting that visible feedback systems could create emotional engagement even in limited interactions.

However, the pilot also revealed a lack of emotional depth, as the short interaction gave players little opportunity to form meaningful connections with the character. As a result, the next iteration introduced additional NPCs, expanded dialogue interactions, and implemented a company boss character to strengthen themes of authority and manipulation.

4.3 Playtest Iteration and Findings

Following the pilot phase, the first formal playtest introduced a more complete version of the game featuring a functional office environment, a fully implemented "Guilt-o-meter" system, a boss character, and two additional employees with whom players could interact. A total of seven participants took part in the playtesting process. Players explored the environment for approximately five to ten minutes while gameplay sessions were recorded, followed by semi-structured interviews.

The findings revealed differing emotional responses among participants. Some players reported that the "Guilt-o-meter" strongly influenced how they interpreted characters during conversations. One participant explained that they initially focused almost entirely on the meter before gradually forming their own opinions through dialogue interactions. Another participant noted that the mechanic subtly guided their interpretation of the characters' intentions, making it difficult to remain emotionally neutral during conversations.

These responses suggest that the “Guilt-o-meter” successfully shaped player perception and influenced moral judgment, supporting the study’s hypothesis that emotional feedback systems can affect player decision-making. Rather than acting as a passive interface element, the mechanic became part of the player’s reasoning process. This supports Gibbons' (n.d.) discussion of moral choice systems, where game mechanics become part of players' ethical decision-making processes.

However, player emotional engagement with the NPCs varied significantly. Some participants described the employees as feeling like real people whose outcomes mattered, while others viewed them primarily as gameplay obstacles rather than emotionally developed characters. One participant stated that the interactions were too short to establish a strong emotional connection, which reduced the emotional impact of the firing decisions.

The results also demonstrated differences in how players experienced moral discomfort. One participant reported feeling uncomfortable while simultaneously experiencing a sense of power when deciding whether to fire employees. In contrast, another participant stated that they did not feel uncomfortable because the situations did not become emotionally intense enough. A third participant dismissed the emotional weight of the decisions entirely, explaining that “it’s just a game.”

These findings suggest that immersion and emotional attachment play a significant role in whether players experience guilt during moral decision-making. This aligns with Liu's (2025) argument that emotional engagement strengthens the impact of moral choices in games. Players who are emotionally connected with the NPCs appeared more likely to hesitate or reflect on their choices, while players who remained detached viewed the interactions more mechanically.

4.4 Technical Challenges and Limitations

Although the dialogue system, office environment, and “Guilt-o-meter” were implemented, technical issues and limited development time prevented the game from including its intended ending. The planned conclusion would have revealed that the company manipulated the investigation to target the highest-paid employee, challenging the player’s trust in the system and encouraging reflection on their earlier decisions.

Without this ending, the research could not fully examine how players might respond to the consequences of their actions. However, the playtests still showed that emotional feedback

systems influenced player decision-making and created moments of discomfort, hesitation, and moral reflection. These findings suggest that mechanics such as the “Guilt-o-meter” have strong potential to increase emotional engagement in narrative-driven games, especially when paired with deeper character development and meaningful consequences.

5 Discussion & Conclusion

This study explored how narrative-driven choices and emotional feedback systems influence players' moral composure, particularly feelings of guilt and moral reflection. Through the development and testing of a prototype office investigation game, the findings suggest that emotional systems such as the "Guilt-o-meter" can significantly influence how players interpret characters and make moral decisions during gameplay. Although the game prototype was not fully completed, the results still provide valuable insight into how affective game design can shape player experience.

One of the most important findings was that the "Guilt-o-meter" affected how players perceived the NPCs and their actions. This finding aligns with Gibbons' (n.d.) historical overview of moral choice systems in games, which highlights how game mechanics can guide player interpretation and influence ethical judgments. Like the systems described by Gibbons, the "Guilt-o-meter" did not directly force decisions but shaped the context in which those decisions were made. Several participants reported that the mechanic guided their judgment during conversations, especially during the early stages of gameplay. This supports theories of affective game design, which argue that emotional feedback systems can deepen immersion and alter player interpretation within interactive narratives. Rather than simply functioning as a gameplay mechanic, the "Guilt-o-meter" became part of the player's moral reasoning process. This demonstrates how interface design itself can influence emotional engagement and ethical decision-making in games.

The findings also highlight the importance of immersion and emotional attachment in creating feelings of guilt. Participants who viewed the NPCs as believable individuals were more likely to experience hesitation, discomfort, or responsibility when deciding whether to fire them. In contrast, players who remained emotionally detached often treated the interactions as mechanical systems rather than moral dilemmas. This aligns with previous research on moral decision-making in games, which suggests that emotional investment is necessary for players to experience ethical conflict within virtual environments. Similarly, Liu (2025) suggests that moral choices become more meaningful when players are emotionally invested in the game world.

An unexpected finding was that some participants trusted the "Guilt-o-meter" more than their own judgment, even when they recognized that the system might not be fully reliable. This

suggests that players may naturally rely on authoritative systems presented by games, especially when uncertainty is intentionally created. This has broader implications for game design, as it demonstrates how mechanics can manipulate player trust and shape moral outcomes without explicit narrative pressure. This is consistent with Gibbons' (n.d.) discussion of how game systems can subtly guide player behavior and judgment.

Through the development process, an important design lesson emerged regarding the relationship between emotional impact and narrative depth. While the “Guilt-o-meter” successfully influenced player perception, the short dialogue interactions limited the emotional attachment players formed with the NPCs. As a result, some participants did not experience strong guilt because the consequences lacked personal weight. This suggests that emotional systems alone are insufficient without meaningful narrative development and stronger character relationships.

The study also faced several limitations. Technical difficulties and limited development time prevented the implementation of the planned ending sequence and a second round of playtesting. Because of this, the research could not fully examine how players might react to long-term narrative consequences or revelations about the company’s manipulation. Additionally, the sample size limits the generalizability of the findings.

Despite these limitations, the findings contribute to the growing body of research on moral decision-making and affective narrative design in games. Liu (2025) suggests that interactive moral choices provide opportunities for players to evaluate their own values through gameplay experiences. The reactions observed during playtesting indicate that even a relatively simple emotional feedback mechanic can encourage players to reconsider their assumptions, question authority, and reflect on the consequences of their actions. This study contributes to affective narrative design research by introducing the “Guilt-o-meter” as a mechanic that visualizes emotional judgment within gameplay. The findings suggest that emotional feedback systems can encourage reflection, influence player perception, and deepen immersion when combined with morally ambiguous narratives. Future research could expand on this work by implementing a complete narrative structure, increasing playtesting samples, and exploring how long-term consequences affect player guilt and moral reflection over extended gameplay experiences.

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6.1 Ludography

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7 Appendix (A)

Interview Questions

Q: What are your thoughts about the dialogue presented during the playtest?

Q: What are your thoughts about the dialogue choices presented during the playtest?

Q: What are your thoughts about the overall narrative presented during the playtest?

Q: Did the Guilt-o-Meter influence how you felt about the people you were talking to or did you form your own opinion through the conversations?

Q: Did the conversations with the NPCs make you feel anything towards them or did they feel like obstacles to get through?

Q: At any point during the conversations did you feel uncomfortable about what you were being asked to do even knowing it's just a game?

Q: What impression, if any, did you get of the character you play as?

Q: When speaking to the characters in the game did you make choices from the perspective of your player character or did you play as yourself?